Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

int *a = b.

0 [arr] = 1%4 = 1.

1 [arr] = 2%4 = 2.

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=4;i++)
4       (i-1)[arr] = (i%4);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 2;
8   }
9  int main(){
10   int arr[5] = {1,1,1,1,1};
11   func1(arr+1);
12   for(int i=0; i<5; i++)
13          printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<5; i++)
17          printf("%d ", i[arr] );
18   return 0;
19 }
```

arr | 1 | 1 | 1 | 1 | 1 |

arr | 1 | 1 | 2 | 3 | 0 |

int *arr = arr + 1

(0) [arr]

a[1] = arr. int arr = *arr

a[0] =

Output
1 1 2 3 0
3 3 4 5 2.
Will the answer change if we write func2 as: **void func2 (int a[1] , int arr){** **int temp = arr + 2;** **a = &temp;** **}** Explain your answer
No.

Q. 2: Write the output of the following sequence of instructions **long arr2[60];**
int arr1[50], a1=(&arr2[40]), a2=(&arr2[30]), b1=(&arr1[40]), b2=(&arr1[30]);
printf("%d %d", a1-a2 , b1-b2); Explain your answer. (2+2 Marks)

10 10

the variable a1 stores the number which corresponds to the address of the 40th element of the array arr2 and its value it 10 greater than the value of a2 thus their difference is 10. likewise for b1 & b2.